

CHE 264 – Chemical Engineering Process and Project Laboratory Spring 2026

Description: The overarching course objective is for students to integrate, practice, and extend the material covered in their core Chemical Engineering courses through experimentation and the development of research projects. Students acquire hands-on experience in unit operations, experimenting in small- and pilot-scale chemical process equipment. Students will gain expertise designing experiments and solving open-ended problems, and they are expected to work successfully in teams as they collect, analyze, and interpret data. Moreover, students are required to propose solutions to scale-up and design problems based on their experimental results. As they complete this work, students learn to prepare written laboratory reports, technical memos, and proposals, and to prepare and deliver oral presentations.

Learning Objectives: By the completion of this course, students will be able to:

- Design experimental plans and perform laboratory experiments logically.
- Apply critical thinking, theoretical models, and chemical engineering principles to analyze experimental data.
- Verify experimental deviations from theoretical models utilizing chemical engineering principles.
- Evaluate the direct application of engineering concepts in the solution of real-world problems.
- Summarize experimental findings, create compelling discussions, and propose practical engineering solutions.
- Develop engineering knowledge by researching and experimenting.

By the practice of these objectives, students will be able to:

- Improve their technical communication skills
- Demonstrate and practice laboratory safety and process safety
- Work effectively in teams while enhancing their individual teaming skills

Materials: Canvas will be used to share course resources. Laboratory manuals and other information will be posted there, as will links to other sources of information.

Lecture: M 8:00-9:00 am in CPE 2.212 or T 11:00 am -12:00 pm in RLP 1.106 on the dates listed in this document.

Laboratory: M, T, W, or Th, 3:00-7:00 pm. Main entrance to laboratory in CPE 1.480 (Laboratory Room Numbers CPE 1.420, 1.440, and 1.480)

Instructor: Professor Carlos Landaverde-Alvarado

Office hours:

- **Professor Landaverde:** Questions? Please stop by the lab on M, T, W from 4:00 pm to 5:00 pm, or you can email me for an appointment if those times don't work for you.
- **TAs:** Please note office hours are mandatory on the week following each laboratory experiment at 5:00 pm. Otherwise, please email the TAs to set a time for a meeting between 4:30 pm and 5:00 pm Monday through Thursday. They will be available on days the laboratory is running as listed in the laboratory schedule.

Grades:

Your course grade will be based on three components:

- 60% Standard Experiments. There are three standard laboratory experiments (Reaction, Distillation, and GSM), which will be given equal weight. Components for each experiment include pre-lab work and a final report.
- 30.5% Team Experimental Project. Each team will complete a Class Project. Successful project completion includes one individual project pre-proposal, a final team written project proposal and presentation, and a poster presentation of the project results.
- 6.5% Participation and Laboratory Safety. Active engagement in planning and conducting the laboratory experiments and analyzing the results is a critical part of this course. Laboratory safety is a very important part of your professional development as a chemical engineer. All group members are expected to fully participate and to follow the safety rules of the laboratory. Participation and safety will be judged primarily by the instructor and TAs. However, you will be required to fill out a team peer evaluation twice during the course of the semester. In addition, your attendance and engagement in the weekly lectures will also count towards this grade.
- 3% Completion of Project Safety Assignment. All student teams must participate in a safety assignment required for the completion of their final project.

The full range of available grades (including + and -) will be utilized. Grades may be curved as necessary.

A	A-	B+	B	B-	C+	C	C-	D	F
93.00-100%	90.00-92.99%	87.00-89.99%	83.00-86.99%	80.00-82.99%	77.00-79.99%	73.00-76.99%	70.00-72.99%	60.00-69.99%	<60.00%

Participation and Laboratory Safety:

Experiential Learning is a critical component of this course. It is for this reason that a portion of the entire grade is based upon how effectively you participate in the laboratory experience. In general, it is expected that you will be fully engaged in the planning and running of the experiment. This also includes following the appropriate safety procedures and rules while completing laboratory experiments.

- Attendance and active engagement in all scheduled lectures are required. A lecture schedule is included below.
- You must attend all of your group's laboratory sessions. All students are expected to attend laboratory sessions (for both standard lab sessions and the class project) at least **5 times** during the semester. University-recognized excuses are required if you are unable to do so. University-recognized excuses include medical issues requiring a doctor visit and/or hospitalization, religious holy days, military service, and qualifying medical or military family leave. Please request a class absence notification from Student Emergency Services in these cases. Requests can be submitted [here](#). Job interviews, extracurricular activities, etc. are not approved excuses.
- The best way to ensure active participation is through preparation. You will need to study the laboratory manual and have done sample calculations and the safety section before the lab period so that you know exactly what you are going to do in the lab. You need to have planned out what needs to be done in the lab before the lab period.
- During at least one of the standard experiments, you will be assigned the role of **team leader** or **safety officer**. Details on the exact dates your group is scheduled to come to lab can be found in Canvas under the folder "Syllabus and Schedules". This document also lists your roles as group leader or safety officer during lab experiments.
 - *The group leader is expected to take a leadership role for the group during their assigned week. For example, the leader will contact the TAs and instructor with any questions, schedule meetings, assign tasks, balance the work schedule in the group, and determine best courses of action to accomplish group objectives.*

- *The safety officer is expected to research and understand all safety precautions necessary for the experiment and laboratory. The safety officer should present the safety section of the pre-lab and should also work on the safety sections of all necessary reports, in addition to other tasks assigned by the team leader.*
- *Regardless of your role in a particular lab, each team member is expected to share equally in the responsibilities and effort required to complete the lab.*
- You cannot let one or more group members dominate the group. Everyone needs to speak up about what needs to be done, what is the next step, etc. The instructor and TAs may ask questions to try to encourage participation by all group members. However, it is each of your responsibility to make sure that you actively contribute.
- Peer evaluations will also represent a percentage of your final participation grade and will play a role in your final course grade. Students with consistently low participation grades will be contacted by the instructors throughout the semester to pursue strategies to improve their participation and engagement in the course.
- Your final participation grade will also be calculated based on your attendance to our weekly lectures. Each student can miss one lecture in the semester without presenting documentation to justify this absence. This policy only applies to lectures, it does not apply to other class activities.

Course Organization:

- Experiments will be performed in groups (designated RED, YELLOW, ORANGE, BLUE, PURPLE and GREEN groups). Groups are assigned by the instructor.
- Over the course of the semester, each group will do each of the 3 standard experiments and a final class project.
- In addition to the 3 laboratory sessions to complete the standard experiments, the class project **will require at least 2 extra mandatory** sessions of in-lab work.
- In-lab standard experiments begin Tuesday, January 20 and conclude Monday, March 2.
- Laboratory time to complete class projects will be scheduled from Monday, March 23 until Thursday, April 9.
- Project poster presentations will be held from Monday, April 20 to Thursday, April 23.
- Course lecture schedule:

Date	Topics
January 12 (M-W labs) January 13 (T-Th labs)	Introductory Lecture
January 26 (M-W labs) January 27 (T-Th labs)	Value Creation Class Projects I
February 2 (M-W labs) February 3 (T-Th labs)	Oral Presentations and Scientific Talks
February 9 (M-W labs) February 10 (T-Th labs)	Technical Memos Teamwork and Teaming Skills
February 16 (M-W labs) February 17 (T-Th labs)	Class Projects II Feedback and Peer-review The Team Contract
February 23 (M-W labs) February 24 (T-Th labs)	Class Project Discussion Safety Course Feedback
April 13 (M-W labs) April 14 (T-Th labs)	Posters and Poster Presentations

Laboratory Safety:

- Long pants and closed-toe shoes are required. You must also wear a lab coat and safety glasses, which will be supplied by the department.
- You must wear disposal gloves, supplied by the department, whenever actively handling chemicals.
- You must enter the laboratory through the marked entrance in room 1.480. For your reference, this is the entrance door across from the elevators.

Laboratory Deadlines: all deadlines listed under assignments in Canvas

Deadline	Individual/Group	Task
Standard Experiments		
5 pm on the week of January 12 on your lab day	Group	Mandatory laboratory tours.
3 pm the day your group is to run an experiment	Group	Submit pre-lab report to Canvas.
3 pm on the day your group is to run an experiment	Group	Complete pre-lab presentation before each experiment.
5:00 pm one week after your team runs an experiment	Group	Attend required office hours
11:59 pm one week after your group runs an experiment	Group	Submit Report Appendix .
11:59 pm two weeks after your group runs an experiment	Group	Submit a draft of the report Results & Discussions section.
11:59 pm on dates listed on the laboratory schedule	Group	Submit final report.
Peer-Review Activities		
11:59 pm on March 13	Individual	Complete 1 st peer evaluation.
11:59 pm on April 24	Individual	Complete 2 nd peer evaluation.
11:59 pm on February 20, February 27, and March 13	Individual	Complete proposal peer-review assignments
Class Project		
11:59 pm on January 16	Group	Submit class project selection (top 2 preferences)
11:59 pm on February 6	Individual	Submit individual project preproposal.
TBA week of February 23 and March 2	Group	Mandatory meetings to discuss class projects.
11:59 pm on March 6	Group	1. Submit written project proposal 2. Submit list of laboratory supplies for class projects
11:59 pm on March 6	Group	Deadline to schedule Safety meetings with ExxonMobil technologists
After 3:00 pm week of March 9	Group	Complete proposal presentation
11:59 pm on March 13	Group	Submit signed Team Contract

11:59 pm on March 27	Group	Deadline to complete Safety meetings with ExxonMobil technologists
4:00 pm on the week of April 20	Group	Poster Session and Project Safety Summary
Note: Posters must be printed 2 business days in advance: For example, if Monday session is on April 20 at 4:00 pm, poster must be submitted by Thursday, April 16 at 4:00 pm. Tuesday session posters must be submitted by Friday, April 17 at noon.		

Late work policy: All laboratory work must be submitted to Canvas by the assigned due date. Late work will have 25% deducted for every 24 hours past the original due date/time. Assignments received more than 48 hours past the original due date will not be accepted.

- The only exception to this policy is the case of a University-recognized excuse (where we receive a notification from Student Emergency Services). University-recognized excuses include medical issues requiring a doctor visit and/or hospitalization, religious holy days, military service, and qualifying medical or military family leave. Please request a class absence notification from Student Emergency Services [here](#).

Regrade Requests: You have **one week** after a report is returned to you (i.e., posted to Canvas) to request a re-grade. Regrading requests should be submitted in writing by email to the experiment TA. They should consist of a few sentences describing your rationale for why a re-grade is necessary for each report section.

Communication: The course Canvas site can be found at utexas.instructure.com. Please email me through Canvas. You are responsible for ensuring that the primary email address you have recorded with the university is the one you will check for the course. This is the email address that Canvas uses.

University Policies and Resources: For a list of important university policies and helpful resources that you may need as you engage with and navigate your courses and the university, see the [University Policies and Resources Students Canvas](#) page. The page includes the language of the University Honor Code, Title IX legal requirements for Texas employees, and information about how to receive support through the office of Disability & Access.

Academic Integrity: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the University is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community. The University of Texas has an honor code, and you are expected to abide by it. You may find a copy of it at: <http://deanofstudents.utexas.edu/conduct/>

Using someone else's work in your own writing without giving proper credit is considered plagiarism, a serious form of academic dishonesty that can result in severe penalties. Copying someone else's work, buying a paper and submitting it as your own, copying and pasting text (even with changes), or borrowing images from an online source, are some examples of plagiarism. Even if you plagiarize accidentally, you can be held responsible and penalized.

In this course, a basic expectation is that the work you turn in is your own and that your actions demonstrate respect for yourself, your classmates, teaching assistants, and the instructors. Your written work should represent *your* understanding of the experiment. This means that copying all or part of another student's (or group) laboratory report is not acceptable, nor is giving a friend your completed laboratory report so that they may copy it. Please do not hesitate to ask questions if you have any doubts as to whether any action or behavior on your part or the part of others constitutes academic dishonest.

Learning to cite sources appropriately is an important part of becoming a professional. When you are unsure about citation, you are encouraged to **ask us** what is appropriate in the context of your assignment. Consultants at The [University Writing Center](#) can also help you determine whether you are citing sources correctly—and they have helpful guides online for using [direct quotations](#) and [paraphrasing](#). Reviewing those skills will help you feel confident that you are handling sources professionally in your writing. You can read the University's definition of plagiarism and other forms of academic dishonesty in Sec. 11–402 of the [Student Conduct Code](#). For more information, visit the [Dean of Students' site](#). All assignments in this course may be processed by TurnItIn, a tool that compares submitted material to an archived database of published work to check for potential plagiarism. Other methods may also be used to determine if a paper is the student's original work. Regardless of the results of any TurnItIn submission, the faculty member will make the final determination as to whether or not a paper has been plagiarized.

No use of generative AI tools permitted: This course assumes that work submitted by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment, and/or using generative AI tools (such as ChatGPT).

Students with Disabilities: The university is committed to creating an accessible and inclusive learning environment for students with disabilities consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A's website for contact and more information: <http://disability.utexas.edu/>. If you are already registered with D&A , please deliver your accommodation letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have our explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. We are well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.